PAYMENT SERVICE — COVERAGE & NOTES

Test execution: 09 Nov 2026 | Maven: clean verify | Java 25.0.3 | Spring Boot 4.1.1

Result: BUILD SUCCESS — 161 tests passed; 0 failures, 0 errors, 0 skipped. **Spring contexts:** 6 **Containers:** 5× PostgreSQL 18, 1× Kafka 4.1.1

TEST BREAKDOWN

Category	Tests	Scope
Unit	72	Isolated application / domain logic
Web MVC slice	7	PaymentControllerWebMvcTest; Mock Servlet / DispatcherServlet
Integration	16	Consumer, outbox releaser, payment creation and payment service
Container-focused	7	Consumed-event storage and Liquibase/payment storage
E2E	59	XML/JSON REST and asynchronous processing / retrieval
Total	**161**	

SPRING CONTEXTS & CONTAINERS

# / Context	Tests using the context	Context / infrastructure
1. Consumer integration	PaymentEventConsumerServiceIntegrationTest (3); ConsumedEventStorageJpaContainerTest (3)	Full @SpringBootTest, local profile, JPA/Hibernate, Liquibase, Hikari; PostgreSQL 18 Testcontainer. Kafka listener and outbox scheduler disabled in shared integration config.
2. Outbox releaser	PaymentEventReleaserServiceIntegrationTest (3)	Full @SpringBootTest, local profile, JPA/Hibernate, Liquibase, Hikari; PostgreSQL 18 Testcontainer; publisher mocked.
3. Payment MVC slice	PaymentControllerWebMvcTest (7)	Spring MVC test slice, Mock Servlet / DispatcherServlet; no database container.
4. Payment persistence/service	CreatePaymentPain001ServiceIntegrationTest (6); PaymentServiceIntegrationTest (4); PaymentStorageContainerLiquibaseTest (4)	Full @SpringBootTest, local profile, JPA/Hibernate, Liquibase, Hikari; PostgreSQL 18 Testcontainer.
5. Async E2E processing	Pain001XmlE2EProcessingTest (2); PaymentJsonE2EProcessingTest (2)	Full web context, random-port Tomcat, Kafka listener active; PostgreSQL 18 + Apache Kafka 4.1.1 Testcontainers.
6. REST E2E	Pain001XmlE2ETest (26); PaymentJsonE2ETest (26); RetrievePaymentE2ETest (3)	Full web context, random-port Tomcat; PostgreSQL 18 Testcontainer; no Kafka container in this context.

JACOCO COVERAGE SUMMARY

Metric	Covered	Missed	Total	Coverage
Lines	544	16	560	**97.14%**
Instructions	2025	109	2134	**94.89%**
Branches	78	20	98	**79.59%**
Methods	115	8	123	**93.50%**
Complexity	144	28	172	**83.72%**

Line coverage: 97.14%. The Maven JaCoCo rule configured at 94% is satisfied (544/560 lines covered; margin +3.14 percentage points). CSV aggregate across 32 classes.

NOTES

- Test categories are mutually exclusive in the 161-test breakdown. Testcontainers are also used by integration and E2E tests; “Container-focused” is a separate reporting bucket, not the total number of tests that use containers.
- Container count: 5 PostgreSQL 18 containers across the integration/E2E contexts and 1 Apache Kafka 4.1.1 container; Testcontainers Ryuk helper containers are excluded.
- Spring context count is based on six context startup events in the supplied Maven log. Contexts are reused by the listed test classes.
- JaCoCo CSV provides aggregate coverage for 32 classes. Percentages are calculated from covered/missed counters; the 94% Maven gate refers to the configured line-coverage threshold.